

Rossmann Store Sales Prediction Using Neural Networks

Subject: M507 – Methods of Prediction

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1 Problem Statement

Accurate sales forecasting is critical for retail organisations to support inventory planning, workforce allocation, and promotional decision-making. The objective of this assignment is to predict daily sales for Rossmann stores using historical sales data combined with store-level attributes. Since sales values are continuous, the task is formulated as a supervised regression problem.

2 Solution Approach

A machine learning approach based on feed-forward neural networks was adopted. First, a baseline neural network was implemented to establish reference performance. Subsequently, an improved neural network incorporating increased capacity, dropout regularisation, and early stopping was developed to enhance predictive accuracy and generalisation. Model performance was evaluated using unseen data.

3 Data Preparation and Splitting

The Rossmann dataset was merged with store metadata to include additional explanatory variables. Records corresponding to closed stores and zero sales were removed. Categorical variables were transformed using one-hot encoding, and numerical features were standardised. To avoid information leakage, the dataset was sorted chronologically before splitting.

Table 1: Dataset Split Proportions

Subset	Percentage of Data
Training Set	80%
Validation Set	10%
Test Set	5%
Hold-out Set	5%

4 Model Architecture

4.1 Baseline Neural Network

The baseline model consists of a simple feed-forward neural network with a single hidden layer. This model serves as a reference point and highlights the limitations of a basic archi-

texture when applied to complex retail sales data.

4.2 Improved Neural Network

The improved model includes multiple hidden layers and dropout regularisation. Dropout randomly deactivates neurons during training, reducing overfitting and improving generalisation (Srivastava et al., 2014).

5 Early Stopping

Early stopping was applied to the improved neural network by monitoring validation loss (`val_loss`) with a patience of five epochs. Training was halted once validation performance ceased to improve, and the best-performing model weights were restored. This approach prevents overfitting while retaining the optimal model state (Goodfellow et al., 2016).

6 Evaluation Metrics

Model performance was evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). MAE measures the average absolute deviation between predicted and actual values, while RMSE penalises larger errors more strongly.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

7 Learning Curve Analysis

7.1 Baseline Model

Figures 1 and 2 show gradual improvement but a persistent gap between training and validation curves, indicating underfitting and limited model capacity.

7.2 Improved Model

The improved model demonstrates rapid convergence and close alignment between training and validation curves, indicating strong generalisation and effective regularisation.

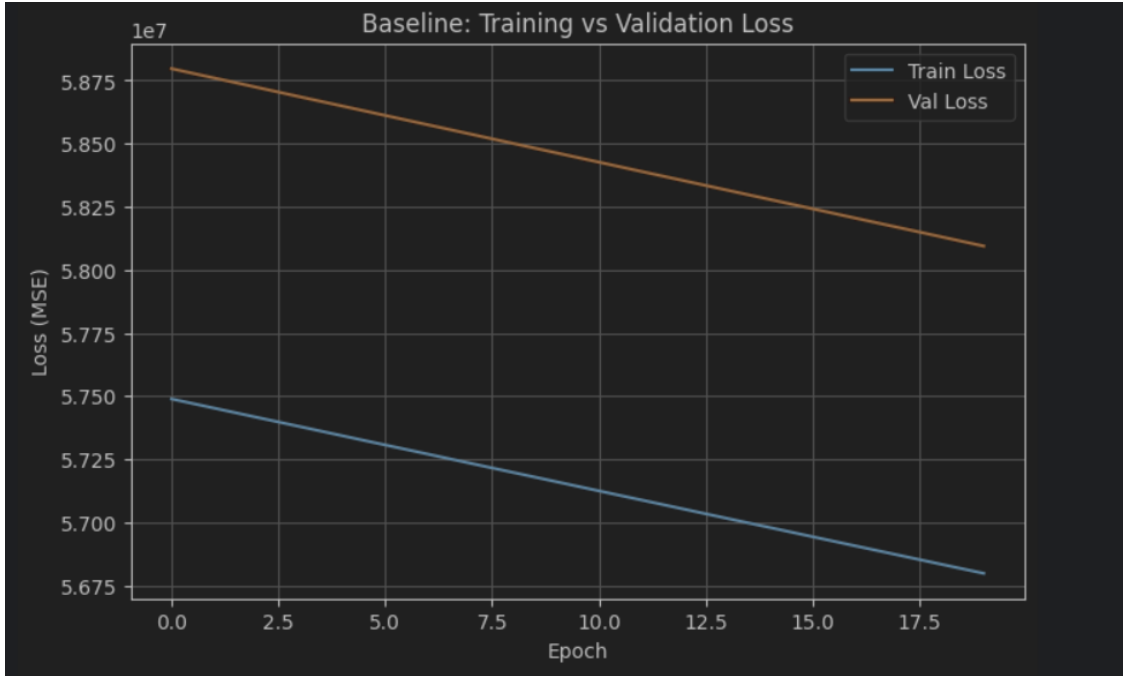


Figure 1: Baseline model: Training and validation loss (MSE) over epochs.

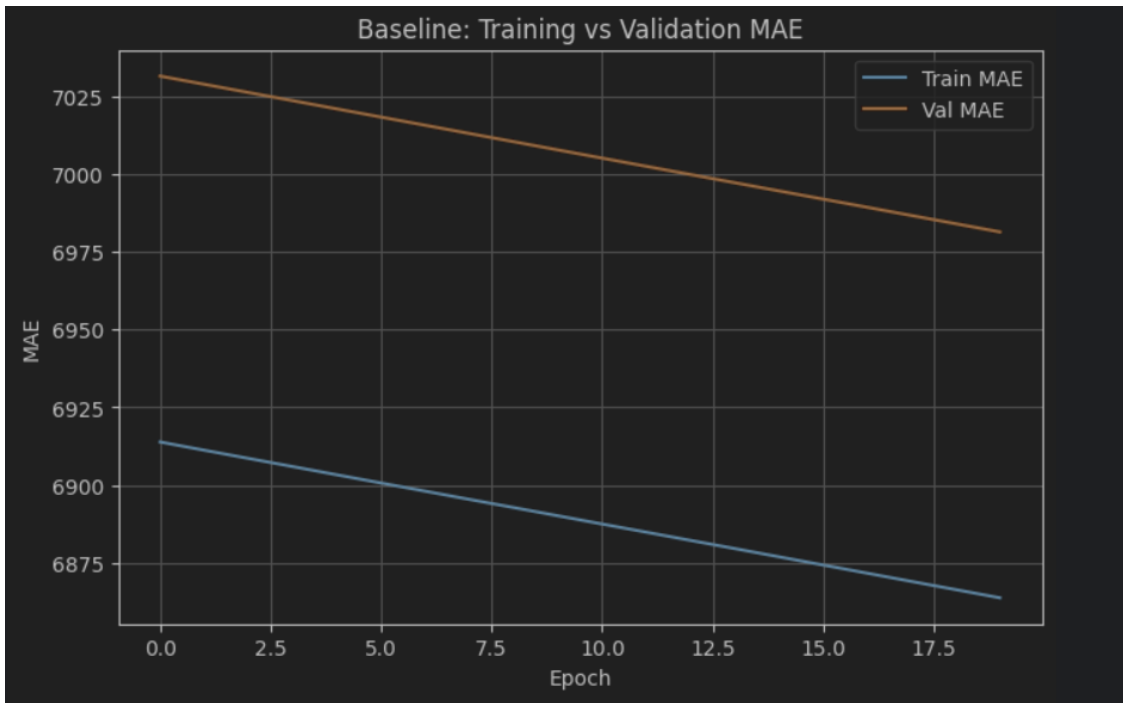


Figure 2: Baseline model: Training and validation MAE over epochs.

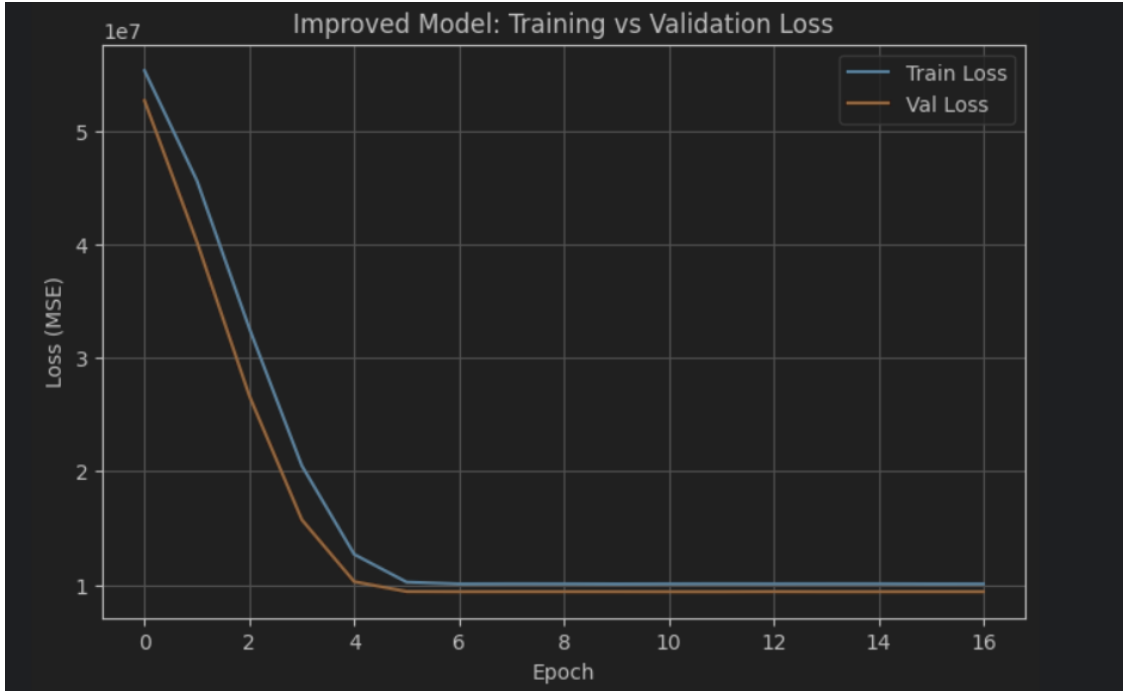


Figure 3: Improved model: Training and validation loss (MSE) over epochs.

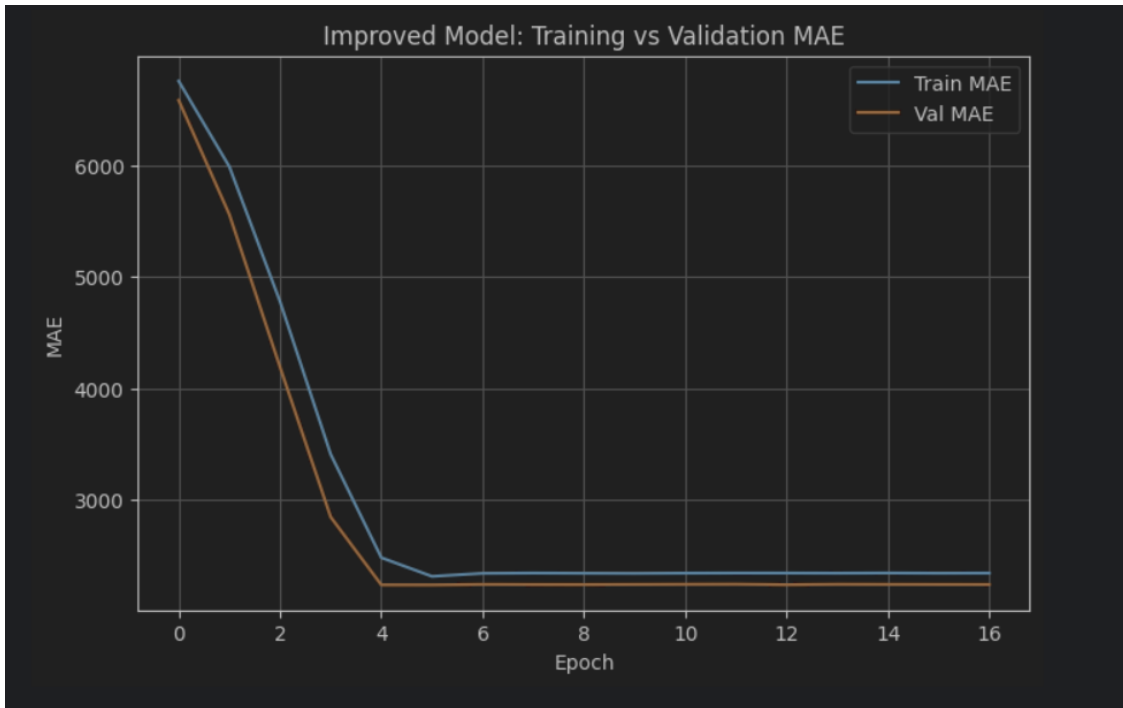


Figure 4: Improved model: Training and validation MAE over epochs.

8 Results

The improved model achieves approximately a 60% reduction in prediction error compared to the baseline across unseen datasets.

Table 2: Model Performance Comparison

Model	Test MAE	Test RMSE	Hold-out RMSE
Baseline Neural Network	7354.32	7974.73	7581.76
Improved Neural Network	2235.52	3127.88	3037.62

9 Conclusion

This study successfully addressed the Rossmann store sales forecasting problem using neural network models. The improved model significantly outperformed the baseline through the use of dropout regularisation and early stopping, demonstrating strong predictive accuracy and generalisation. The results fulfil the objectives of the M507 Methods of Prediction module.

Tools and Environment

Model development and experimentation were carried out using Python. Data exploration and preliminary analysis were supported using Microsoft Excel. The development environment included IntelliJ IDEA and DataSpell for code editing and experimentation. This report was typeset using the L^AT_EX code editor and compiled using the Overleaf online platform.

References

Goodfellow, I., Bengio, Y. and Courville, A. (2016) *Deep Learning*. Cambridge, MA: MIT Press.

Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I. and Salakhutdinov, R. (2014) ‘Dropout: A simple way to prevent neural networks from overfitting’, *Journal of Machine Learning Research*, 15(1), pp. 1929–1958.

Rossmann (2015) *Rossmann Store Sales Dataset*. Available at: <https://www.kaggle.com/c/rossmann-store-sales> (Accessed: 2025).